

Tasmanian devil cathelicidins exhibit anticancer activity against Devil Facial Tumour Disease (DFTD) cells

Keywords: transcriptomics, transmissible cancer, antimicrobial peptides, enrichment analysis.

Link to the paper

<https://www.nature.com/articles/s41598-023-39901-0>

Project Aim

The aim of this project is to understand and reproduce selected figures from the chosen paper. In the first step, each member of the group will reproduce the same figure, which should result from a relatively simpler analysis. Subsequently, participants will choose another figure from the provided list and reproduce it. Bonus points will be awarded for reproducing additional figures.

The project's expectations include reproducing a main figure and supplementary figure(s) from the paper listed below. A short report should be written, comprising:

- Brief Introduction: Provide background information and motivation for the paper.
- Methods: Explain the main steps followed to reproduce the figures.
- Figures and Discussion: Analyse the obtained figures. Could you reproduce the figures exactly as they appear in the paper? Justify your answer.
- Conclusion

Paper Details

- The Tasmanian devil (*Sarcophilus harrisii*) is endangered due to the spread of Devil Facial Tumour Disease (DFTD), a contagious cancer with no current treatment options.
 - DFTD is known to be able to transmit between devils due to their lack of major histocompatibility complex class I (MHC-I) molecules, which helps it evade host immune responses.
- While a myriad of treatment options have been explored, from medication to vaccinations, none have progressed to clinical trials.
- This paper seeks to assess the potential anticancer properties of Tasmanian devil's own cathelicidins. Cathelicidins are a major class of vertebrate antimicrobial peptides, and while are archetypally 'antimicrobial', they are pleiotropic in nature, with some cathelicidins from some species exhibiting immunomodulatory or even anticancer activity.
- To test their efficacy, a cytotoxicity assay was run on DFT1 cell line 1426 against seven Tasmanian devil cathelicidins at 12, 18, 24 and 36h. RNAseq analysis was further performed to understand the mechanisms of action of these cathelicidins on the DFT1 cell line.
- Four Tasmanian devil cathelicidins (Saha-CATH3, 4, 5 and 6) decreased cell viability by more than 50% over 36h compared to the growth control at 500 µg/mL.
- RNAseq revealed that these cathelicidins generated more differentially expressed genes than the non-toxic cathelicidins (Saha-CATH1, 2 and 7).
- Performing GO analysis on these four cathelicidins revealed downregulation of genes involved in DNA replication, cell cycle progression and checkpoints.
 - In particular, Saha-CATH5 also regulated the ERBB and Hippo signaling pathways in DFT1 cells. It is known that DFTD cells are sensitive to Receptor Tyrosine Kinase (RTK) inhibitors in vitro, and the oncogenic activity of RTK signaling pathways is often amplified by the formation of a positive feedback loop with the Hippo pathway. It is likely that Saha-CATH5 is acting under a similar mechanism.

The data

All necessary data to reproduce the figures of interest, such as the counts matrix, will provided to you.

Main figure to reproduce

Using a counts matrix provided to you, replicate Figures 2 and 3. Following the publication, use edgeR to normalise the gene counts matrix, perform DGE using the limma package, and run over-representation analysis (ORA) using clusterProfiler. Provide a brief explanation of each computational step (comments in your code will suffice).

Suggested supplementary figures and analysis

Perform at least one of the following analysis. You may reproduce more for bonus points:

1. **Volcano plot:** for each cathelicidin against the control, generate a volcano plot visualising the differentially expressed genes. Use the same FDR and LFC cutoffs as described in the publication. Do your plots match with the results in the publication?
2. **GSEA:** Instead of performing overrepresentation analysis, use the GSEA software to determine which gene sets are enriched in the cathelicidin groups. Present the top results for Saha-CATH3, 4, 5 and 6 - are they the same as the ORA? Why or why not?
 1. Hint: First generate a GSEA.rnk file (a list of the genes ranked from largest to smallest log fold change).
3. **Methods comparison:** This publication outlines a pipeline to run pathway enrichment analysis, which goes from g:profiler, GSEA and Cytoscape <https://www.nature.com/articles/s41596-018-0103-9>. Run through the pipeline outlined in this paper, and present your results. How does this method differ from the approach used in the publication? What are the pros and cons of both?

Project Organization

Every Tuesday from 5pm to 7pm, I will be available to answer all project-related questions. On other days, you can reach me via email if necessary (simon.tang@epfl.ch).

During the first session, I will explain the project to the whole group, describe the data, and then allow you to start reproducing the figures.

Note that the project is individual, even though the first figure is common to all members.